

APPENDIX E

STORAGE OF AIRCRAFT

APPENDIX OVERVIEW

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E-1 INTRODUCTION AND SCOPE.

E-1.1 Introduction.

This task contains instructions for storage of helicopter and activation after storage. Storage of helicopter includes corrosion control, which consists primarily of using preservatives to prevent moisture from contacting exposed metal surfaces. There are two main types of corrosion, direct chemical and electrochemical, each is defined as follows:

(1) Direct chemical corrosion: Occurs when chemicals in the atmosphere cause metal surfaces to erode or etch.

(2) Electrochemical corrosion: Occurs when adjacent surfaces of dissimilar metals become electrolytic cells in the presence of moisture.

Preservation procedures are based on the principle that clean, moisture-free surfaces will not corrode. Preservation of the helicopter consists of cleaning the exposed metal surfaces and providing a barrier between the surfaces and the moisture-laden atmosphere.

E-1.2 Environmental Conditions.

- a. The weather must be taken into account when a helicopter is to be placed in storage. The category of storage selected will determine the amount of ground operation, motoring of engines, and other required maintenance for which men and materials are needed. Wet weather conditions create corrosion, rot, mildew, and mold. To prevent environmental conditions from damaging equipment, perform inspections regularly and take proper preventive maintenance action.

- b. The following practices should be used as a guide during exceptionally wet weather conditions:

- (1) Prevent rot, mildew, and mold by keeping fabrics, rubber, and other susceptible materials as dry as possible. Keep fabric material in the helicopter clean.
- (2) Treat all visible corrosion.
- (3) Wash helicopter often.
- (4) Lubricate moving parts when helicopter is exposed to salt-laden air or mud.
- (5) Open all drain holes to make sure water drains properly.
- (6) If tanks are not preserved, keep fuel tanks full to prevent condensation in tanks.

E-1.3 Categories of Storage.

The information in this task does not cover when or why helicopters are stored and activated, only how this is done. The instructions in this task cover three types of helicopter storage. The three types and their duration are:

FLYABLE STORAGE	NO TIME LIMIT
SHORT-TERM STORAGE	1 TO 45 DAYS
INTERMEDIATE STORAGE	46 TO 180 DAYS

E-2 FLYABLE, SHORT-TERM, AND INTERMEDIATE STORAGE.

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INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Fan
- Thermometer

References

- MIL-STD-794
- PARA 1-3-3

- PARA 1-3-5
- PARA 1-5-1 through PARA 1-5-13
- PARA 1-6-2
- PARA 1-6-4
- PARA 1-6-6
- SEI 671
- TM 1-2840-248-23

E-2.1 General Procedures.

- Use the following procedures whenever a helicopter is placed in a flyable, short-term, or intermediate storage:
- Tow helicopter (PARA 1-6-2).
 - Park helicopter (PARA 1-6-6).
 - Observe fire regulations for helicopters parked outside.
 - Tie down helicopter (PARA 1-6-4).
 - Lubricate helicopter before placing it in storage (PARA 1-5-1 through PARA 1-5-13).
 - Protect engines from corrosion resulting from dirt, moisture, and disuse (SEI 671 and TM 1-2840-248-23).
 - Cover and protect elastomeric bearings from sunlight.

E-2.2 Inspection During Storage.

- The local maintenance officer is responsible for maintaining proper helicopter ventilation and preservation; maintaining helicopter drain valves in operable condition and free from obstructions (open drain valves for the duration of the storage period); providing drains in protective covers at locations where water collects; and for establishing an inspection program for stored helicopters. Inspect as follows:
- Determine peak interior helicopter temperatures during hot weather conditions as follows:
 - Hang standard thermometers temporarily in helicopter.
 - Record interior temperatures at 30-minute intervals during hottest part of day.

NOTE

If normal ventilating methods are not enough, provide fan ventilation.

(3) If interior temperatures go over 160°F (71°C), ventilate helicopter.

- b. Periodically inspect helicopter for corrosion. Frequency of inspection will be determined by local commander based on humidity, temperature, atmospheric conditions and storage conditions. When found, treat corrosion as required.
- c. Inspect tiedown devices (ground wire, wheel chocks, mooring ropes, rods, and eyes) at regular intervals for bending and signs of wear. Inspect tiedown devices immediately after helicopter has been subjected to winds over 40 mph. If any tiedown device is bent or show signs of wear, replace tiedown device.

- d. Inspect protective covers for proper drainage. If cover(s) are damaged or worn, replace cover.
- e. Inspect communication equipment for fungus and corrosion. If fungus or corrosion is found, perform the following:

NOTE

Do not remove or package antennas.

- (1) Remove, repair, and package communication equipment that is deteriorating (MIL-STD-794).
- (2) Store packaged equipment in helicopter from which it was removed.
- f. Inspect tires for at least 75% normal tire pressure during storage period. If tire pressure is below 75%, service landing gear tire(s) (PARA 1-3-3 or PARA 1-3-5).

E-3 PREPARING HELICOPTER FOR FLYABLE STORAGE.

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INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

Materials/Parts

- Barrier Material, Item 80, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Machinery Towel, Item 211, Appendix D
- Pressure Sensitive Adhesive Tape, Item 245, Appendix D

References

- Appendix D
- Appropriate Flight Manual
- PARA E-2.2
- PARA 1-3-2

- PARA 1-3-3
- PARA 1-3-4
- PARA 1-3-5
- PARA 1-3-8
- PARA 1-3-9
- PARA 1-3-12
- PARA 1-3-13
- PARA 1-3-14
- PARA 1-3-15
- PARA 1-3-18
- PARA 1-6-5
- PARA 9-4-34
- PARA 9-4-106
- SEI 671
- TM 1-2840-248-23

- Appropriate MTF

E-3.1 Inspection Before Flyable Storage.

NOTE

Helicopters in flyable storage will be kept in a serviceable condition.

- Install static ground wire.
- Perform 10-hour engine inspection (SEI 671 and TM 1-2840-248-23).
- Enter in helicopter logbook. Include date and type of storage.

E-3.2 Engines.

- Preserve engines per short-term storage criteria (SEI 671 and TM 1-2840-248-23).
- Install engine exhaust plugs (PARA 1-6-5).
- Install engine inlet plugs (PARA 1-6-5).

E-3.3 Auxiliary Power Unit.

- Check and service APU (PARA 1-3-9).
- Start and operate APU for 10 to 15 minutes (Appropriate Flight Manual).
- Install APU exhaust plug (PARA 1-6-5).
- Seal APU inlet with barrier material, Item 80, Appendix D. Secure with pressure sensitive adhesive tape, Item 245, Appendix D.

E-3.4 Hydraulic Systems.

- Service hydraulic systems (PARA 1-3-8).
- Check accumulators for required pressure (PARA 1-3-8).

E-3.5 Main Rotor Head.

- Check damper indicators on dampers for proper fluid level (PARA 1-3-15).

- If necessary, service dampers (PARA 1-3-15).

E-3.6 Gear Boxes.

If necessary, check and service main transmission (PARA 1-3-12), intermediate gear box (PARA 1-3-13), and tail gear box (PARA 1-3-14) to proper oil level.

E-3.7 Fuel Cells.

- Drain water from fuel cells before adding fuel.

NOTE

Keep fuel cells at full level for duration of storage.

- Refuel main fuel cells (PARA 1-3-18).

E-3.8 Electrical System.

- Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- On helicopters with NiCad battery, disconnect electrical connector, P141, from NiCad battery and connect to dummy electrical connector on battery relay junction box (PARA 9-4-34).
- Wrap battery electrical connector with barrier material, Item 80, Appendix D. Secure with pressure sensitive adhesive tape, Item 245, Appendix D.

NOTE

Remove battery from helicopter when ambient temperature is below -40°F (-40°C). Store battery in heated building.

- If required, remove battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).

E-3.9 Landing Gear.

- Maintain normal tire pressure (PARA 1-3-3 and PARA 1-3-5).

- b. Maintain normal shock strut extension (PARA 1-3-2 and PARA 1-3-4).
- c. Clean polished chrome surfaces on exposed pistons using machinery towel, Item 211, Appendix D, and dry-cleaning solvent, Item 124, Appendix D.
- d. Coat cleaned piston surfaces with hydraulic fluid, Item 154 or Item 155, Appendix D.

E-3.10 Airframe.

- a. Close doors and windows unless ventilation is required.
- b. Cover all fuselage openings to keep water, dust, or other foreign matter out.

E-3.11 Flight Controls.

Release friction lock on pilot's collective stick.

E-3.12 Inspection During Flyable Storage.

- a. Inspect using local directives.
- b. Inspect helicopter (PARA E-2.2).

NOTE

Engine inspection is required every 14 days.

- c. Perform 14-day engine inspection (SEI 671 and TM 1-2840-248-23).

NOTE

- Make sure main and tail rotor blades are turned during engine run up.
- Engine run up is required every 14 days.
- d. Run up engine every 14 days (Appropriate MTF).

E-4 PREPARING HELICOPTER FOR SHORT-TERM STORAGE.

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INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Protection, Eye

Materials/Parts

- Barrier Material, Item 80, Appendix D
- Condition Tag, Green
- Corrosion Preventive Compound, Item 112, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Lubricating Oil, Item 207, Appendix D
- Machinery Towel, Item 211, Appendix D

- Nitrogen, Item 221, Appendix D

- Pressure Sensitive Adhesive Tape, Item 245, Appendix D

- Rags

References

- Appendix D
- PARA E-2.2
- PARA 1-3-2
- PARA 1-3-3
- PARA 1-3-4
- PARA 1-3-5
- PARA 1-3-8
- PARA 1-3-9
- PARA 1-3-10

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- PARA 1-3-12
- PARA 1-3-13
- PARA 1-3-14
- PARA 1-3-15
- PARA 1-4-1
- PARA 1-6-4
- PARA 1-6-5
- PARA 1-6-7
- PARA 9-4-34
- PARA 9-4-106
- SEI 671
- TM 1-2840-248-23
- TM 1-70-PMS-1
- TM 11-70-23

E-4.1 Inspection Before Short-Term Storage.

NOTE

Helicopters in short-term storage will be kept in a serviceable condition.

- a. Install static ground wire.
- b. Perform 10 hour/14 day inspection (TM 1-70-PMS-1).
- c. Enter in helicopter logbook. Include date, type of storage, and removed or disconnected components.

E-4.2 Engines.

- a. Preserve engines (SEI 671 and TM 1-2840-248-23).
- b. Install engine exhaust plugs (PARA 1-6-5).
- c. Install engine inlet plugs (PARA 1-6-5).

E-4.3 Auxiliary Power Unit.

- a. Check and service APU oil tank to FULL (PARA 1-3-9).
- b. Disconnect electrical connector from ignition exciter.
- c. Insulate electrical connector.
- d. Disconnect fuel line from fuel manifold.

- e. Disconnect fuel line from start fuel injector.

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

- f. Blow out remaining fuel from fuel manifold and start fuel injector. Use dry, compressed air or bottled nitrogen, Item 221, Appendix D, at 30 psi.
- g. Place rags under disconnected lines to catch fuel and oil.
- h. Disconnect fuel line from fuel inlet filter.
- i. Connect gravity-feed supply of lubricating oil, Item 207, Appendix D, to filter fitting.
- j. Place APU switch to START and motor APU until APU automatically shuts down.
- k. Recharge accumulator, using external power or APU handpump (PARA 1-3-10).

- l. Repeat motoring until oil flows through fuel control and out of drain.
- m. Disconnect oil supply.
- n. Motor APU to remove any residual preservative oil from fuel system.
- o. Recharge accumulator (PARA 1-3-10).
- p. Disconnect drain lines.
- q. Connect fuel line to fuel inlet filter.
- r. Connect fuel line to fuel manifold.
- s. Connect fuel line to start fuel injector.
- t. Connect electrical connector to ignition exciter.
- u. Place a green condition tag on the APU engine stating: "APU FUEL SYSTEM PRESERVED WITH LUBRICATING OIL, MIL-PRF-6081. FLUSHING REQUIRED BEFORE STARTING."

CAUTION

Do not allow dry-cleaning solvent to come in contact with wire insulation, rubber, or gasket material surfaces. These surfaces may be damaged by dry-cleaning solvent.

- v. Clean APU external surfaces with machinery towel, Item 211, Appendix D, and dry-cleaning solvent, Item 124, Appendix D.

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing

compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

- w. Dry APU with machinery towel, Item 211, Appendix D, and blow dry with compressed air.
- x. Install APU exhaust plug (PARA 1-6-5).
- y. Seal APU inlet with barrier material, Item 80, Appendix D. Secure with pressure sensitive adhesive tape, Item 245, Appendix D.

E-4.4 Hydraulic Systems.

- a. Service hydraulic systems (PARA 1-3-8).
- b. Check accumulator for proper charge (PARA 1-3-8).
- c. Clean exposed pistons of all servos with machinery towel, Item 211, Appendix D, and dry-cleaning solvent, Item 124, Appendix D.
- d. Coat all cleaned pistons with hydraulic fluid, Item 154, Appendix D.

E-4.5 Gear Boxes.

- a. Check, and if necessary, service main transmission (PARA 1-3-12), intermediate gear box (PARA 1-3-13), and tail gear box (PARA 1-3-14) to proper oil level.
- b. Coat exposed surfaces of drive train parts with corrosion preventive compound, Item 112, Appendix D.

E-4.6 Main Rotor Blades.

If necessary, fold main rotor blades (PARA 1-6-7).

E-4.7 Main Rotor Head.

- a. Check damper indicators on dampers for proper fluid level (PARA 1-3-15).
- b. If necessary, service dampers (PARA 1-3-15).



Do not allow corrosion preventative compound to come in contact with bearings or rubber boots. Surfaces may be damaged by corrosion preventative compound.

- c. Coat exposed surfaces with corrosion preventative compound, Item 112, Appendix D.
- d. Cover elastomeric bearings with barrier material, Item 80, Appendix D. Secure with pressure sensitive adhesive tape Item 245, Appendix D, to protect from sunlight.

E-4.8 Tail Rotor.

- a. Coat exposed metal surfaces with corrosion preventive compound, Item 112, Appendix D.
- b. Cover hub, pitch change links, and tail gear box output shaft with barrier material, Item 80, Appendix D. Secure with pressure sensitive adhesive tape Item 245, Appendix D.

E-4.9 Fuel Cells.

- a. Helicopter will be stored with fuel cells at full level for duration of storage.
- b. Drain water from cells before adding fuel.

E-4.10 Electrical System.

- a. Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- b. On helicopters with NiCad battery, disconnect electrical connector, P141, from NiCad battery and connect to dummy electrical connector on battery relay junction box (PARA 9-4-34).
- c. Wrap battery electrical connector with barrier material, Item 80, Appendix D. Secure with pressure sensitive adhesive tape Item 245, Appendix D.

NOTE

Remove battery from helicopter when ambient temperature is below -40°F (-40°C). Store battery in heated building.

- d. If required, remove battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).

E-4.11 Instruments.

- a. Clean pitot tube with dry-cleaning solvent, Item 124, Appendix D.
- b. Install pitot tube cover (PARA 1-6-5).

E-4.12 Avionics.

NOTE

Do not remove unclassified avionics from helicopter unless malfunctions exist. Repaired avionics must be installed as soon as component returns from the repair shop.

- a. Remove, protect, and return classified avionics equipment to proper storage facility as called for in publications that cover these components.
- b. Remove 9-volt battery from Doppler computer display unit (TM 11-70-23).

E-4.13 Landing Gear.

- a. Maintain normal tire pressure (PARA 1-3-3 and PARA 1-3-5).
- b. Maintain normal shock strut extension (PARA 1-3-2 and PARA 1-3-4).
- c. Clean polished chrome surfaces on exposed pistons using machinery towel, Item 211, Appendix D, and dry-cleaning solvent, Item 124, Appendix D.
- d. Coat cleaned piston surfaces with hydraulic fluid, Item 154 or Item 155, Appendix D.
- e. Cover tires with barrier material, Item 80, Appendix D. Secure with pressure sensitive adhesive tape, Item 245, Appendix D.

E-4.14 Airframe.

- a. Clean interior and exterior fuselage surfaces (PARA 1-4-1).
- b. Close doors and windows unless ventilation is required.
- c. Cover all fuselage openings to keep out water, dust, or other foreign matter.
- d. Moor helicopter (PARA 1-6-4).

E-4.15 Flight Controls.

Release friction lock on pilot's collective stick.

E-4.16 Inspection During Short-Term Storage.

- a. Inspect using local directives.
- b. Inspect helicopter (PARA E-2.2).

E-5 PREPARING HELICOPTER FOR INTERMEDIATE STORAGE.

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INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Protection, Eye

Materials/Parts

- Barrier Material, Item 80, Appendix D
- Condition Tag, Green
- Corrosion Preventive Compound, Item 112, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Lubricating Oil, Item 207, Appendix D

- Machinery Towel, Item 211, Appendix D
- Nitrogen, Item 221, Appendix D
- Pressure Sensitive Adhesive Tape, Item 245, Appendix D
- Rags
- Tags

References

- Appendix D
- PARA E-2.2
- PARA 1-3-2
- PARA 1-3-3
- PARA 1-3-4
- PARA 1-3-5

- PARA 1-3-8
- PARA 1-3-9
- PARA 1-3-10
- PARA 1-3-12
- PARA 1-3-13
- PARA 1-3-14
- PARA 1-3-15
- PARA 1-4-1
- PARA 1-4-5
- PARA 1-6-4
- PARA 1-6-5
- PARA 5-4-31
- PARA 5-4-38
- PARA 8-4-28
- PARA 9-4-34
- PARA 9-4-106
- PARA 10-4-40
- PARA 17-4-1
- PARA 17-4-2
- SEI 671
- TM 1-2840-248-23
- TM 1-70-PMS-1
- TM 11-70-23

E-5.1 Inspection Before Intermediate Storage.

NOTE

Helicopters in intermediate storage will be kept in a serviceable condition.

- a. Install static ground wire.
- b. Perform 10 hour/14 day inspection (TM 1-70-PMS-1).
- c. Enter in helicopter logbook. Include date, type of storage, and removed or disconnected components.

E-5.2 Engines.

- a. Preserve engines (SEI 671 and TM 1-2840-248-23).

- b. Install engine exhaust plugs (PARA 1-6-5).
- c. Install engine inlet plugs (PARA 1-6-5).

E-5.3 Auxiliary Power Unit.

- a. Check and service APU oil tank to FULL (PARA 1-3-9).
- b. Disconnect electrical connector from ignition exciter.
- c. Insulate electrical connector.
- d. Disconnect fuel line from fuel manifold.
- e. Disconnect fuel line from start fuel injector.

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

- f. Blow out remaining fuel from fuel manifold and start fuel injector. Use dry, compressed air or bottled nitrogen, Item 221, Appendix D, at 30 psi.
- g. Place rags under disconnected lines to catch fuel and oil.
- h. Disconnect fuel line from fuel inlet filter.
- i. Connect gravity-feed supply of lubricating oil, Item 207, Appendix D, to filter fitting.
- j. Place APU switch to START and motor APU until APU automatically shuts down.
- k. Recharge accumulator, using external power or APU handpump (PARA 1-3-10).
- l. Repeat motoring until oil flows through fuel control and out of drain.
- m. Disconnect oil supply.
- n. Motor APU to remove any residual preservative oil from fuel system.
- o. Recharge accumulator (PARA 1-3-10).
- p. Disconnect drain lines.
- q. Connect fuel line to fuel inlet filter.
- r. Connect fuel line to fuel manifold.
- s. Connect fuel line to start fuel injector.

- t. Connect electrical connector to ignition exciter.
- u. Place a green condition tag on the APU engine stating: "APU FUEL SYSTEM PRESERVED WITH LUBRICATING OIL, MIL-PRF-6081. FLUSHING REQUIRED BEFORE STARTING."

CAUTION

Do not allow dry-cleaning solvent to come in contact with wire insulation, rubber, or gasket material surfaces. Surfaces may be damaged by dry-cleaning solvent.

- v. Clean APU external surfaces with machinery towel, Item 211, Appendix D, and dry-cleaning solvent, Item 124, Appendix D.

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

- w. Dry APU with machinery towel, Item 211, Appendix D, then blow dry with compressed air.
- x. Install APU exhaust plug (PARA 1-6-5).
- y. Seal APU inlet with barrier material, Item 80, Appendix D. Secure with pressure sensitive adhesive tape, Item 245, Appendix D.

E-5.4 Hydraulic Systems.

- a. Service hydraulic systems (PARA 1-3-8).
- b. Check accumulator for proper charge (PARA 1-3-8).

- c. Clean exposed pistons of all servos with machinery towel, Item 211, Appendix D, and dry-cleaning solvent, Item 124, Appendix D.
- d. Coat all cleaned pistons with hydraulic fluid, Item 154 or Item 155, Appendix D.

E-5.5 Gear Boxes.

- a. Check, and if necessary, service main transmission (PARA 1-3-12), intermediate gear box (PARA 1-3-13), and tail gear box (PARA 1-3-14) to proper oil level.
- b. Coat exposed surfaces of drive train components with corrosion preventive compound, Item 112, Appendix D.

E-5.6 Main Rotor Blades.

- a. Remove main rotor blades (PARA 5-4-31).
- b. Tag main rotor blades with helicopter serial number.
- c. Clean main rotor blade cuffs (PARA 1-4-5).
- d. Coat main rotor blade pin sockets with corrosion preventive compound, Item 112, Appendix D.
- e. Cover main rotor blade cuffs with barrier material, Item 80, Appendix D. Secure with pressure sensitive adhesive tape, Item 245, Appendix D.
- f. Clean main rotor blades (PARA 1-4-5).
- g. Place main rotor blades in shipping containers.

E-5.7 Main Rotor Head.

- a. Check damper indicators on dampers for proper fluid level (PARA 1-3-15).
- b. If necessary, service dampers (PARA 1-3-15).

CAUTION

Do not allow corrosion preventative compound to come in contact with bear-

ings or rubber boots. Surfaces may be damaged by corrosion preventative compound.

- c. Coat exposed surfaces with corrosion preventive compound, Item 112, Appendix D.
- d. Cover elastomeric bearings with barrier material, Item 80, Appendix D. Secure with pressure sensitive adhesive tape, Item 245, Appendix D, to protect from sunlight.

E-5.8 Tail Rotor.

- a. Remove tail rotor blades (PARA 5-4-38).
- b. Tag and store tail rotor components (other than tail rotor blades) in cabin area.
- c. Tag tail rotor blades with helicopter serial number and place in shipping containers.
- d. Place tail rotor blades in shipping containers.
- e. Coat exposed metal surfaces of tail rotor with corrosion preventive compound, Item 112, Appendix D.
- f. Cover tail gear box output shaft with barrier material, Item 80, Appendix D. Secure with pressure sensitive adhesive tape, Item 245, Appendix D.

E-5.9 Fuel Cells.

Helicopter will be stored with fuel cells drained and cells purged (PARA 10-4-40).

E-5.10 Electrical System.

- a. Disconnect electrical connector, P140, from battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- b. On helicopters with NiCad battery, disconnect electrical connector, P141, from NiCad battery and connect to dummy electrical connector on battery relay junction box (PARA 9-4-34).
- c. Wrap battery electrical connector with barrier material, Item 80, Appendix D. Secure with pressure sensitive adhesive tape, Item 245, Appendix D.

- d. Remove battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)). Transfer battery to battery shop or other heated building.

E-5.11 Instruments.

- a. Remove clocks (PARA 8-4-28).
- b. Attach a green condition tag on clock and store in secure facility.
- c. Clean pitot tube with dry-cleaning solvent, Item 124, Appendix D.
- d. Install pitot tube cover (PARA 1-6-5).

E-5.12 Avionics.

NOTE

Do not remove unclassified communication equipment from helicopter unless malfunctions exist. Repaired communication equipment must be installed immediately upon return from the repair shop.

- a. Remove, protect, and return classified communication equipment to proper storage facility as noted in applicable directives.
- b. Cover disconnected connectors with barrier material, Item 80, Appendix D. Secure with pressure sensitive adhesive tape, Item 245, Appendix D.
- c. Clean antenna masts, metal mounts, brackets, and associated mechanical items. Use dry-cleaning solvent, Item 124, Appendix D.
- d. Coat cleaned surfaces. Use corrosion preventive compound, Item 112, Appendix D.
- e. If necessary, remove mechanical items showing evidence of corrosion.
- f. Remove corrosion, preserve, package, and store mechanical items with equipment from which it was removed.
- g. Remove 9-volt battery from Doppler computer display unit (TM 11-70-23).

E-5.13 Landing Gear.

- a. Maintain normal tire pressure (PARA 1-3-3 and PARA 1-3-5).
- b. Maintain normal shock strut extension (PARA 1-3-2 and PARA 1-3-4).
- c. Clean polished chrome surfaces on exposed pistons using machinery towel, Item 211, Appendix D, and dry-cleaning solvent, Item 124, Appendix D.
- d. Coat cleaned piston surfaces with hydraulic fluid, Item 154 or Item 155, Appendix D.
- e. Cover tires with barrier material, Item 80, Appendix D. Secure with pressure sensitive adhesive tape, Item 245, Appendix D.

E-5.14 Utility Items.

- a. Remove fire extinguishers from cockpit and cabin (PARA 17-4-1 and PARA 17-4-2).
- b. Transfer fire extinguishers to proper storage facility.
- c. Remove first aid kits (PARA 17-4-1 and PARA 17-4-2).

E-5.15 Airframe.

- a. Clean interior and exterior fuselage surfaces (PARA 1-4-1).
- b. Close doors and windows unless ventilation is required.
- c. Cover fuselage openings to keep out water, dust, or other foreign matter.
- d. Clean external surfaces of plastic windows.
- e. Dry windows and surrounding areas thoroughly.
- f. Cover windows with barrier material, Item 80, Appendix D. Secure with pressure sensitive adhesive tape, Item 245, Appendix D.
- g. Moor helicopter (PARA 1-6-4).

E-5.16 Flight Controls.

- b. Inspect helicopter (PARA E-2.2).

Release friction lock on pilot's collective stick.

E-5.17 Inspection During Intermediate Storage.

- a. Inspect helicopter using local directives.

E-6 PUTTING HELICOPTER INTO SERVICE AFTER FLYABLE STORAGE.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
E-6.1	Airframe	E-20	E-6.7	Hydraulic Systems	E-20
E-6.2	Landing Gear	E-20	E-6.8	Engines	E-20
E-6.3	Electrical System	E-20	E-6.9	Auxiliary Power Unit	E-20
E-6.4	Fuel Cells	E-20	E-6.10	Inspection After Activation From Flyable Storage	E-21
E-6.5	Gear Boxes	E-20			
E-6.6	Main Rotor Head	E-20			

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

Materials/Parts

- Dry-Cleaning Solvent, Item 124, Appendix D
- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Machinery Towel, Item 211, Appendix D

References

- Appendix D
- Appropriate Flight Manual
- PARA 1-3-2
- PARA 1-3-3
- PARA 1-3-4
- PARA 1-3-5

- PARA 1-3-8
- PARA 1-3-9
- PARA 1-3-12
- PARA 1-3-13
- PARA 1-3-14
- PARA 1-3-15
- PARA 1-3-17
- PARA 1-4-1
- PARA 1-6-4
- PARA 1-6-5
- PARA 9-4-34
- PARA 9-4-106
- SEI 671
- TM 1-2840-248-23
- TM 1-70-PMS-1

E-6.1 Airframe.

- a. Before performing operational checks, make sure all systems and components are depressured and serviced.
- b. Remove tiedown restraints (PARA 1-6-4).
- c. Remove protective covers and stow (PARA 1-6-5).
- d. Remove all barrier material.
- e. Clean residue left by barrier material tape with dry-cleaning solvent, Item 124, Appendix D.
- f. Clean interior and exterior fuselage surfaces (PARA 1-4-1).
- g. Open doors and windows.

E-6.2 Landing Gear.

- a. Check tires for proper inflation, condition and rotation (PARA 1-3-3 and PARA 1-3-5).
- b. Clean polished chrome surfaces on exposed pistons using machinery towel, Item 211, Appendix D, and dry-cleaning solvent, Item 124, Appendix D.
- c. Coat cleaned piston surfaces with hydraulic fluid, Item 154 or Item 155, Appendix D.
- d. Service shock struts (PARA 1-3-2 and PARA 1-3-4).

E-6.3 Electrical System.

- a. If required, install battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- b. Remove tape and barrier material from battery connector.
- c. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- d. On helicopters with NiCad battery, connect electrical connector, P141, to NiCad battery (PARA 9-4-34).

E-6.4 Fuel Cells.

- a. Drain water from fuel cells.
- b. Service fuel cells as required (PARA 1-3-17).

E-6.5 Gear Boxes.

If necessary, check and service main transmission (PARA 1-3-12), intermediate gear box (PARA 1-3-13), and tail gear box (PARA 1-3-14) to proper oil level.

E-6.6 Main Rotor Head.

- a. Check damper indicators on dampers for proper fluid level (PARA 1-3-15).
- b. If necessary, service dampers (PARA 1-3-15).

E-6.7 Hydraulic Systems.

- a. Service hydraulic systems (PARA 1-3-8).
- b. Check hydraulic accumulators for required pre-charge pressure (PARA 1-3-8).

E-6.8 Engines.

- a. Remove and stow engine inlet and exhaust plugs (PARA 1-6-5).
- b. Check and service engine (SEI 671 and TM 1-2840-248-23).
- c. Clean engine external surfaces, as necessary (SEI 671 and TM 1-2840-248-23).

E-6.9 Auxiliary Power Unit.

- a. Remove tape and barrier material from intake.
- b. Remove and stow APU exhaust plug (PARA 1-6-5).
- c. Check and service APU (PARA 1-3-9).
- d. Start and operate APU for 5 minutes (Appropriate Flight Manual).
- e. Shut down APU (Appropriate Flight Manual).

E-6.10 Inspection After Activation From Flyable Storage.

- a. Check helicopter logbook for record of components removed or disconnected. Check that components have been installed or connected.
- b. Check that all procedures required to activate helicopter from flyable storage have been done.
- c. Check that helicopter logbook entries have been posted.
- d. Perform 10 hour/14 day inspection (TM 1-70-PMS-1).

E-7 PUTTING HELICOPTER INTO SERVICE AFTER SHORT-TERM STORAGE.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
E-7.1	Airframe	E-23	E-7.9	Main Rotor Blades	E-24
E-7.2	Landing Gear	E-23	E-7.10	Gear Boxes	E-24
E-7.3	Avionics	E-23	E-7.11	Hydraulic Systems	E-24
E-7.4	Instruments	E-23	E-7.12	Auxiliary Power Unit	E-24
E-7.5	Electrical System	E-23	E-7.13	Engines	E-25
E-7.6	Fuel Cells	E-23	E-7.14	Inspection After Activation from Short-Term Storage	E-25
E-7.7	Tail Rotor	E-24			
E-7.8	Main Rotor Head	E-24			

INITIAL SETUP

- PARA 1-3-2

Tools

- PARA 1-3-3

- Aircraft Mechanic’s Toolkit
- Container

- PARA 1-3-4
- PARA 1-3-5

Materials/Parts

- PARA 1-3-8

- Dry-Cleaning Solvent, Item 124, Appendix D
- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Machinery Towel, Item 211, Appendix D
- Tags

- PARA 1-3-9
- PARA 1-3-10
- PARA 1-3-12
- PARA 1-3-13
- PARA 1-3-14

References

- PARA 1-3-15

- Appendix D
- Appropriate Flight Manual

- PARA 1-3-17
- PARA 1-4-1

- PARA 1-6-4
- PARA 1-6-5
- PARA 1-6-7
- PARA 9-4-34
- PARA 9-4-106
- SEI 671
- TM 1-2840-248-23
- TM 1-70-PMS-1
- TM 11-70-23

E-7.1 Airframe.

- a. Before performing operational checks, make sure related systems or components are depre-served and serviced.
- b. Remove tiedown restraints (PARA 1-6-4).
- c. Remove protective covers and stow (PARA 1-6-5).
- d. Remove all barrier material from airframe and components.
- e. Clean residue left by barrier material tape with dry-cleaning solvent, Item 124, Appendix D.
- f. Clean interior and exterior fuselage surfaces (PARA 1-4-1).
- g. Open doors and windows.

E-7.2 Landing Gear.

- a. Check tires for proper inflation, condition and rotation (PARA 1-3-3 and PARA 1-3-5).
- b. Clean polished chrome surfaces on exposed pistons using machinery towel, Item 211, Ap-pendix D, and dry-cleaning solvent, Item 124, Appendix D.
- c. Coat cleaned piston surfaces with hydraulic fluid, Item 154 or Item 155, Appendix D.
- d. Service shock struts (PARA 1-3-2 and PARA 1-3-4).

E-7.3 Avionics.

- a. Requisition classified avionics equipment from supply office. Protection and installation of

classified equipment must be per applicable directives.

- b. Remove tags from avionics equipment.
- c. Install 9-volt battery from Doppler computer display unit (TM 11-70-23).

E-7.4 Instruments.

- a. Remove pitot tube protective cover and stow (PARA 1-6-5).
- b. Remove tape and barrier material sealing static and sideslip ports.
- c. Remove tape residue with dry-cleaning solvent, Item 124, Appendix D.

E-7.5 Electrical System.

- a. If required, install battery (PARA 9-4-34 (Ni-Cad) or PARA 9-4-106 (SLAB)).
- b. Remove tape and barrier material from battery connector.
- c. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- d. On helicopters with NiCad battery, connect electrical connector, P141, to NiCad battery (PARA 9-4-34).

E-7.6 Fuel Cells.

- a. Drain any water from fuel cells.
- b. Service fuel cells as required (PARA 1-3-17).

E-7.7 Tail Rotor.

Clean corrosion preventive compound from tail rotor using machinery towel, Item 211, Appendix D, and dry-cleaning solvent, Item 124, Appendix D.

E-7.8 Main Rotor Head.

CAUTION

Do not allow dry-cleaning solvent to come in contact with elastomeric bearings or rubber boots. These surfaces may be damaged by dry-cleaning solvent.

- a. Clean corrosion preventive compound from main rotor head using machinery towel, Item 211, Appendix D, and dry-cleaning solvent, Item 124, Appendix D.
- b. Check damper indicators on dampers for proper fluid level (PARA 1-3-15).
- c. If necessary, service dampers (PARA 1-3-15).

E-7.9 Main Rotor Blades.

If necessary, unfold main rotor blades (PARA 1-6-7).

E-7.10 Gear Boxes.

- a. Drain some oil and check main transmission for proper oil level (PARA 1-3-12).
- b. Drain some oil and check intermediate and tail gear boxes for proper oil level. Discontinue draining at proper oil level (PARA 1-3-13 and PARA 1-3-14).
- c. Clean corrosion preventive compound from drive train components. Use machinery towel, Item 211, Appendix D, and dry-cleaning solvent, Item 124, Appendix D.

E-7.11 Hydraulic Systems.

- a. Service hydraulic systems (PARA 1-3-8).

- b. Check accumulator for proper pressure (PARA 1-3-8).
- c. Clean exposed pistons of all servos with machinery towel, Item 211, Appendix D, and dry-cleaning solvent, Item 124, Appendix D.
- d. Coat all clean piston surfaces with hydraulic fluid, Item 154 or Item 155, Appendix D.

E-7.12 Auxiliary Power Unit.

- a. Remove tape and barrier material from intake.
- b. Remove and stow APU exhaust plug (PARA 1-6-5).
- c. Check and service APU (PARA 1-3-9).
- d. Disconnect fuel line from fuel manifold.
- e. Disconnect fuel line from fuel injector.
- f. Connect drain lines to disconnected fuel lines.
- g. Place end of drain lines in empty quart container.
- h. Place APU switch to START and motor APU until APU automatically shuts down.
- i. Recharge accumulator, using external power or APU handpump (PARA 1-3-10).
- j. Repeat motoring until oil-free fuel flows through fuel control and out of drain lines.
- k. Recharge accumulator to required pressure (PARA 1-3-10).
 - l. Disconnect drain lines.
- m. Connect fuel line to fuel manifold.
- n. Connect fuel line to start fuel injector.
- o. Connect electrical connector to ignition exciter.
- p. Drain accumulated fuel from combustor fuel drain.
- q. Start and operate APU for 5 minutes (Appropriate Flight Manual).

E-7.13 Engines.

- a. Remove and stow engine inlet and exhaust plugs (PARA 1-6-5).
- b. Depreserve engine (SEI 671 and TM 1-2840-248-23).

E-7.14 Inspection After Activation from Short-Term Storage.

- a. Check helicopter logbook for record of components removed or disconnected. Check that these components have been installed or connected.

- b. Check that all procedures required to activate helicopter after short-term storage have been done.
- c. Check that all PRESERVED tags have been removed.
- d. Check that helicopter logbook entries have been posted for helicopter, engines, and APU.
- e. Perform 10 hour/14 day inspection (TM 1-70-PMS-1).

E-8 PUTTING HELICOPTER INTO SERVICE AFTER INTERMEDIATE STORAGE.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
E-8.1	Airframe	E-27	E-8.10	Main Rotor Blades	E-28
E-8.2	Landing Gear	E-27	E-8.11	Gear Boxes	E-28
E-8.3	Utility Items	E-27	E-8.12	Hydraulic Systems	E-28
E-8.4	Avionics	E-27	E-8.13	Auxiliary Power Unit	E-28
E-8.5	Instruments	E-27	E-8.14	Engines	E-29
E-8.6	Electrical System	E-27	E-8.15	Inspection Following Activation from Intermediate Storage	E-29
E-8.7	Fuel Cells	E-28			
E-8.8	Tail Rotor	E-28			
E-8.9	Main Rotor Head	E-28			

INITIAL SETUP

- PARA 1-3-2

Tools

- PARA 1-3-3

- Aircraft Mechanic's Toolkit
- Container

- PARA 1-3-4
- PARA 1-3-5

Materials/Parts

- Dry-Cleaning Solvent, Item 124, Appendix D
- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Machinery Towel, Item 211, Appendix D
- Tags

- PARA 1-3-8
- PARA 1-3-9
- PARA 1-3-10
- PARA 1-3-12
- PARA 1-3-13
- PARA 1-3-14

References

- Appendix D
- Appropriate Flight Manual

- PARA 1-3-15
- PARA 1-3-17

- PARA 1-4-1
 - PARA 1-6-4
 - PARA 1-6-5
 - PARA 5-4-31
 - PARA 5-4-38
 - PARA 8-4-28
 - PARA 9-4-34
 - PARA 9-4-106
 - PARA 17-4-1
 - PARA 17-4-2
 - SEI 671
 - TM 1-2840-248-23
 - TM 1-70-PMS-1
 - TM 11-70-23
-

E-8.1 Airframe.

- a. Before performing operational checks, make sure related systems or components are depre-served and serviced.
- b. Remove tiedown restraints (PARA 1-6-4).
- c. Remove protective covers and stow (PARA 1-6-5).
- d. Remove all barrier material from airframe and components.
- e. Clean residue left by barrier material tape with dry-cleaning solvent, Item 124, Appendix D.
- f. Clean interior and exterior fuselage surfaces (PARA 1-4-1).
- g. Open doors and windows.

E-8.2 Landing Gear.

- a. Check tires for proper inflation and condition (PARA 1-3-3 and PARA 1-3-5).
- b. Clean polished chrome surfaces on exposed pistons using machinery towel, Item 211, Appendix D, and dry-cleaning solvent, Item 124, Appendix D.
- c. Coat cleaned piston surfaces with hydraulic fluid, Item 154 or Item 155, Appendix D.
- d. Service main landing gear and tail landing gear shock struts (PARA 1-3-2 and PARA 1-3-4).

E-8.3 Utility Items.

- a. Install fire extinguishers in cabin and cockpit (PARA 17-4-1 and PARA 17-4-2).
- b. Install first aid kits in cabin and cockpit (PARA 17-4-1 and PARA 17-4-2).

E-8.4 Avionics.

- a. Requisition classified communication equip-ment from supply office. Protect and install classified equipment using applicable direc-tives.
- b. Remove tags from communication equipment.
- c. Install 9-volt battery from Doppler computer display unit (TM 11-70-23).

E-8.5 Instruments.

- a. Remove pitot tube protective cover and stow (PARA 1-6-5).
- b. Remove tape and barrier material sealing static and sideslip ports.
- c. Remove tape residue with dry-cleaning solvent Item 124, Appendix D.
- d. Install clocks (PARA 8-4-28).

E-8.6 Electrical System.

- a. Install battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).

- b. Remove tape and barrier material from battery connector.
- c. Connect electrical connector, P140, to battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
- d. On helicopters with NiCad battery, connect electrical connector, P141, to NiCad battery (PARA 9-4-34).

E-8.7 Fuel Cells.

- a. Prepare fuel cells for service.
- b. Service fuel cells (PARA 1-3-17).

E-8.8 Tail Rotor.

- a. Clean corrosion preventive compound from tail rotor using machinery towel, Item 211, Appendix D, and dry-cleaning solvent, Item 124, Appendix D.
- b. Install tail rotor blades (PARA 5-4-38).

E-8.9 Main Rotor Head.



Do not allow dry-cleaning solvent to come in contact with elastomeric bearings or rubber boots. These surfaces may be damaged by dry-cleaning solvent.

- a. Clean corrosion preventive compound from main rotor head using machinery towel, Item 211, Appendix D, and dry-cleaning solvent, Item 124, Appendix D.
- b. Check damper indicators on dampers for proper fluid level (PARA 1-3-15).
- c. If necessary, service dampers (PARA 1-3-15).

E-8.10 Main Rotor Blades.



Make sure color codes on main rotor blades agree with color code on sleeves.

Install main rotor blades (PARA 5-4-31).

E-8.11 Gear Boxes.

- a. Drain some oil and check main transmission for proper oil level (PARA 1-3-12).
- b. Drain some oil and check intermediate and tail gear boxes for proper oil level. Stop draining at proper oil level (PARA 1-3-13 and PARA 1-3-14).
- c. Clean corrosion preventive compound from drive train components using machinery towel, Item 211, Appendix D, and dry-cleaning solvent, Item 124, Appendix D.

E-8.12 Hydraulic Systems.

- a. Service hydraulic systems (PARA 1-3-8).
- b. Check accumulator for proper pressure (PARA 1-3-8).
- c. Clean exposed pistons of all servos with machinery towel, Item 211, Appendix D, and dry-cleaning solvent, Item 124, Appendix D.
- d. Coat all cleaned pistons with hydraulic fluid, Item 154 or Item 155, Appendix D.

E-8.13 Auxiliary Power Unit.

- a. Remove tape and barrier material from intake.
- b. Remove and store APU exhaust plug (PARA 1-6-5).
- c. Check and service APU (PARA 1-3-9).

- d. Disconnect fuel line from fuel manifold.
- e. Disconnect fuel line from start fuel injector.
- f. Connect drain lines to disconnected fuel lines.
- g. Place end of drain lines in empty quart container.
- h. Place APU switch to START and motor APU until APU automatically shuts down.
- i. Recharge accumulator, using external power or APU handpump (PARA 1-3-10).
- j. Repeat motoring until oil-free fuel flows through fuel control and out of drain lines.
- k. Recharge accumulator to required pressure (PARA 1-3-10).
- l. Disconnect drain lines.
- m. Connect fuel line to fuel manifold.
- n. Connect fuel line to start fuel injector.
- o. Connect electrical connector to ignition exciter.
- p. Drain accumulated fuel from combustor fuel drain.

- q. Start and operate APU for 5 minutes (Appropriate Flight Manual).

E-8.14 Engines.

- a. Remove and stow engine inlet and exhaust plugs (PARA 1-6-5).
- b. Depreserve engine (SEI 671 and TM 1-2840-248-23).

E-8.15 Inspection Following Activation from Intermediate Storage.

- a. Check helicopter logbook for record of components removed or disconnected. Check that these components have been installed or connected.
- b. Check that all procedures required to activate helicopter from intermediate storage have been done.
- c. Check that all PRESERVED tags have been removed.
- d. Check that helicopter logbook entries have been posted for helicopter, engines, and APU.
- e. Perform 10 hour/14 day inspection (TM 1-70-PMS-1).

